Servo Motor

Category: **Electrical & Electronic Components & Kits** Type: **Short SOP** Priority: **Medium** 1 labs

Covers these lab items: Servo Motor

What it is used for

A servo motor provides controlled angular or linear movement in compatible automation and robotics systems. Confirm the motor's electrical and mechanical ratings and configure its controller before operation.

How to use

1. Inspect the motor body, shaft, connector, mounting points, and cable for damage.
2. Check the datasheet for voltage, current, torque, travel limits, and control signal requirements.
3. Mount the motor securely and connect the load without misalignment or binding.
4. Connect the motor to a compatible controller and specified power supply.
5. Set the required control parameters and verify the movement range without load where appropriate.
6. Apply power and test movement at low speed or within a safe range.
7. Switch off power before adjusting the mounting, load, or wiring.

⚠ Safety

- Do not exceed rated voltage, current, torque, or travel limits.
- Keep fingers and loose clothing away from the rotating shaft and connected mechanism.
- Secure the load to prevent unexpected movement.
- Disconnect power before maintenance or changing connections.

Care & storage

- Keep the motor clean, inspect cables and mounting bolts, and follow the manufacturer's lubrication guidance.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.